

Page 4	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2010	9702	41

7 (a) arrow pointing up the page B1 [1]

(b) (i) $Eq = Bqv$ C1
 $v = (12 \times 10^3) / (930 \times 10^{-6})$ C1
 $= 1.3 \times 10^7 \text{ m s}^{-1}$ A1 [3]

(ii) $Bqv = mv^2 / r$ C1
 $q/m = (1.3 \times 10^7) / (7.9 \times 10^{-2} \times 930 \times 10^{-6})$ C1
 $= 1.8 \times 10^{11} \text{ C kg}^{-1}$ A1 [3]

8 (a) momentum conservation hence momenta of photons are equal (but opposite) M1